

实验一

一、实验名称

用 Hermite 分段三次线性插值绘制 `T`, `F`, `5`

二、实验原理

1 由于单变量函数不能描述复杂的曲线图形，我们考虑引入参数 $t \in [0,1]$ ，分别求 $x(t), y(t)$ 的插值函数。在根据 $(x(t), y(t))$ ， $t \in [0,1]$ 绘制曲线。

2 Hermite 分段三次线性插值公式

$$\begin{aligned} H_{2n+1}(t) &= H_{1,0}f(t_0) + H_{1,1}f(t_1) + \hat{H}_{1,0}f'(t_0) + \hat{H}_{1,1}f'(t_1) \\ H_{n,j}(t) &= [1 - 2(t - t_j) \cdot L'_{n,j}(t)] \cdot L_{n,j}^2(t) \\ \hat{H}_{n,j} &= (t - t_j) \cdot L_{n,j}^2(t) \end{aligned}$$

2.1 n 代表插值的节点数-1, (n 从 0 开始);

2.2 $L_{n,j}$ 代表，在 n 节点的拉格朗日插值下，第 j 节点的拉格朗日插值基；

带入 $n=1$

$$\begin{aligned} H_3(t) &= H_{1,0}f(t_0) + H_{1,1}f(t_1) + \hat{H}_{1,0}f'(t_0) + \hat{H}_{1,1}f'(t_1) \\ H_{1,0}(t) &= [1 - 2(t - t_0) \cdot L'_{1,0}(t)] \cdot L_{1,0}^2(t) \\ H_{1,1}(t) &= [1 - 2(t - t_1) \cdot L'_{1,1}(t)] \cdot L_{1,1}^2(t) \\ \hat{H}_{1,0} &= (t - t_0) \cdot L_{1,0}^2(t) \\ \hat{H}_{1,1} &= (t - t_1) \cdot L_{1,1}^2(t) \\ L_{1,0}(t) &= \frac{t - t_1}{t_0 - t_1} \\ L_{1,1}(t) &= \frac{t - t_0}{t_1 - t_0} \end{aligned}$$

3. 初始条件为(以第一段插值为例)

$$\begin{aligned} t_0 &= 0, & t_1 &= 1 \\ x(t_0) &= x_0, x(t_1) = x_1, & x'(t_0) &= 3\alpha_0, x'(t_1) = 3\alpha_1 \\ y(t_0) &= y_0, y(t_1) = y_1, & y'(t_0) &= 3\beta_0, y'(t_1) = 3\beta_1 \end{aligned}$$

带入可得

$$\begin{aligned} H_3(t) &= H_{1,0}x_0 + H_{1,1}x_1 + \hat{H}_{1,0}\alpha_0 + \hat{H}_{1,1}\alpha_1 \\ H_{1,0}(t) &= [1 - 2(t) \cdot (-1)] \cdot (t - 1)^2 \\ H_{1,1}(t) &= [1 - 2(t - 1) \cdot (1)] \cdot t^2 \\ \hat{H}_{1,0} &= (t) \cdot (t - 1)^2 \\ \hat{H}_{1,1} &= (t - 1) \cdot t^2 \end{aligned}$$

化简可得

$$\begin{aligned} x_3(t) &= (2(x_0 - x_1) + (\alpha_0 + \alpha_1)) \cdot t^3 + (3(x_1 - x_0) - (2\alpha_0 + \alpha_1)) \cdot t^2 + t\alpha_0 \\ y_3(t) &= (2(y_0 - y_1) + (\beta_0 + \beta_1)) \cdot t^3 + (3(y_1 - y_0) - (2\alpha_0 + \beta_1)) \cdot t^2 + t\beta_0 + y_0 \end{aligned}$$

为方便计算，再令

$$\begin{aligned} x'_0 &= x_0 + \alpha_0, x'_1 = x_1 - \alpha_1 \\ y'_0 &= y_0 + \beta_0, y'_1 = y_1 - \beta_1 \end{aligned}$$

(x'_0, x'_1, y'_0, y'_1 为guidepoint)

三、算法描述

Input: $n; (x_0, y_0) \cdots (x_n, y_n); (x_0^+, y_0^+) \cdots (x_{n-1}^+, y_{n+1}^+); (x_1^-, y_1^-) \cdots (x_n^+, y_n^+)$

Output: 系数 $\{a_0^i, a_1^i, a_2^i, a_3^i, b_0^i, b_1^i, b_2^i, b_3^i, \text{for } i \text{ in } 1 \rightarrow n-1\}$

Step 1 for each $i = 0, 1, 2, \dots, n-1$ do step 2 and 3

Step 2

$$\begin{aligned} a_0^{(1)} &= x_i; & b_0^{(1)} &= y_i; \\ a_1^{(1)} &= 3(x_i^+ - x_i); & b_1^{(1)} &= 3(y_i^+ - y_i); \\ a_2^{(1)} &= 3(x_1 + x_{i+1}^- - 2x_i^-); & b_2^{(1)} &= 3(y_1 + y_{i+1}^- - 2y_i^-); \\ a_3^{(1)} &= x_{i+1} - x_i + 3x_i^+ - 3x_{i+1}^-; & b_3^{(1)} &= y_{i+1} - y_i + 3y_i^+ - 3y_{i+1}^-; \end{aligned}$$

四、实验数据

五的数据

```
149 597 149 597 149 597 345 597
345 597 361 597 365 599 368 606
368 606 406 695 368 606 406 695
406 695 397 702 406 695 397 702
397 702 382 681 372 676 351 676
351 676 351 676 351 676 142 676
142 676 33 439 142 676 33 439
33 439 32 438 32 436 32 434
32 434 32 428 35 426 44 426
44 426 74 426 109 420 149 408
149 408 269 372 324 310 324 208
324 208 324 112 264 37 185 37
185 37 165 37 149 44 119 66
119 66 86 90 65 99 42 99
42 99 14 99 0 87 0 62
0 62 0 24 46 0 121 0
121 0 205 0 282 27 333 78
333 78 378 123 399 180 399 256
399 256 399 327 381 372 333 422
333 422 288 468 232 491 112 512
112 512 112 512 149 597 149 597
```

f 的数据

```
1 289 452 289 452 166 452 166 452
2 166 452 166 452 166 568 166 568
3 166 568 166 627 185 657 223 657
4 223 657 245 657 258 647 276 618
5 276 618 292 589 304 580 321 580
6 321 580 345 580 363 598 363 621
7 363 621 363 657 319 683 259 683
8 259 683 196 683 144 656 118 611
9 118 611 92 566 84 530 83 450
10 83 450 83 450 1 450 1 450
11 1 450 1 450 1 418 1 418
12 1 418 1 418 83 418 83 418
13 83 418 83 418 83 104 83 104
14 83 104 83 31 72 19 0 15
15 0 15 0 15 0 0 0 0
16 0 0 0 0 260 0 260 0
17 260 0 260 0 260 15 260 15
18 260 15 178 18 167 29 167 104
19 167 104 167 104 167 418 167 418
20 167 418 167 418 289 418 289 418
21 289 418 289 418 289 452 289 452
```

T 的数据

```
1 237 620 237 620 237 120 237 120
2 237 120 237 35 226 24 143 19
3 143 19 143 19 143 0 143 0
4 143 0 143 0 435 0 435 0
5 435 0 435 0 435 19 435 19
6 435 19 353 23 339 36 339 109
7 339 109 339 108 339 620 339 620
8 339 620 339 620 393 620 393 620
9 393 620 507 620 529 602 552 492
10 552 492 552 492 576 492 576 492
11 576 492 576 492 570 662 570 662
12 570 662 570 662 6 662 6 662
13 0 492 0 492 24 492 24 492
14 24 492 48 602 71 620 183 620
15 183 620 183 620 237 620 237 620
16 6 662 6 662 0 492 0 492
```

五、实验环境

Windows 10

IDE: CLion

六、实验程序及结果

程序

```
1. #include <windows.h>
2. #include "iostream"
3. #include "fstream"
4. #include "cmath"
```

```

5. using namespace std;
6. const double PI = 3.1415;
7. static HWND sHwnd;
8. static COLORREF redColor=RGB(255,0,0);
9. static COLORREF blueColor=RGB(0,0,255);
10. static COLORREF greenColor=RGB(0,255,0);
11.
12. void SetWindowHandle(HWND hwnd)
13. {
14.     sHwnd=hwnd;
15. }
16.
17. void setPixel(int x, int y, COLORREF& color=redColor)
18. {
19.     if(sHwnd == NULL)
20.     {
21.         MessageBoxA(NULL,"sHwnd was not initialized !","Error",MB_OK|
22.             MB_ICONERROR);
23.         exit(0);
24.     }
25.     HDC hdc=GetDC(sHwnd);
26.     SetPixel(hdc,x,y,color);
27.     ReleaseDC(sHwnd,hdc);
28.     return;
29. }
30. double Bezier(double c0, double c1, double c2, double c3, double t) {
31.     return c3*t*t*t + c2*t*t + c1*t + c0;
32. }
33.
34. void drawCurve(string filename, int len) {
35.
36.     ifstream infile;
37.     infile.open(filename);
38.     int data[8];
39.     for (int i = 0 ; i < len ; i ++) {
40.         double a0, a1, a2, a3;
41.         double b0 ,b1, b2, b3;
42.         for (int i = 0 ; i < 8 ; i ++) {
43.             infile >> data[i];
44.         }
45.
46.         a0 = data[0];

```

```

47.         b0 = data[1];
48.         a1 = 3*(data[2] - data[0]);
49.         b1 = 3*(data[3] - data[1]);
50.         a2 = 3*(data[0] + data[4] - 2*data[2]);
51.         b2 = 3*(data[1] + data[5] - 2*data[3]);
52.         a3 = data[6] - data[0] + 3*(data[2] - data[4]);
53.         b3 = data[7] - data[1] + 3*(data[3] - data[5]);
54.
55.         printf("%lf %lf %lf %lf %lf %lf %lf %lf \n",
56.             a0, a1, a2, a3, b0, b1, b2, b3);
57.
58.         double t = 0;
59.         double dt = 1.0/1000;
60.         while (fabs(t - 1) > 1e-6) {
61.             double dx = Bezier(a0, a1, a2, a3, t);
62.             double dy = Bezier(b0, b1, b2, b3, t);
63.             //setPixel(dx + 100, 742 - dy , blueColor);
64.             if(sHwnd == NULL)
65.             {
66.                 MessageBoxA(NULL,"sHwnd was not initialized !","Error
67.                 ",MB_OK|MB_ICONERROR);
68.                 exit(0);
69.             }
70.             HDC hdc=GetDC(sHwnd);
71.             SetPixel(hdc,dx + 100,742 - dy,blueColor);
72.             ReleaseDC(sHwnd,hdc);
73.             // 742- 为了保证图像不是倒立的
74.             t += dt;
75.         }
76.     }
77. }
78.
79.
80.
81. LRESULT CALLBACK WndProc(HWND hwnd,UINT message,WPARAM wParam,LPARAM
    lParam)
82. {
83.     switch(message)
84.     {
85.         case WM_PAINT:
86.             SetWindowHandle(hwnd);
87.             //drawSin();
88.             drawCurve("D:\\learning\\c\\clion\\t.dat", 16);

```

```

89.         //drawCurve("D:\\learning\\c\\clion\\f.dat", 21);
90.         //drawCurve("D:\\learning\\c\\clion\\5.dat", 21);
91.         break;
92.     case WM_CLOSE: // Failure to call DefWindowProc
93.         break;
94.     case WM_DESTROY:
95.         PostQuitMessage(0);
96.         return 0;
97.     default:
98.         break; // Failure to call DefWindowProc //
99. }
100. return DefWindowProc(hwnd,message,wParam,lParam);
101. }
102.
103. int WINAPI WinMain(HINSTANCE hInstance,HINSTANCE hPrevInstance,LPS
    TR lpCmdLine,int iCmdShow)
104. {
105.     static TCHAR szAppName[] = TEXT("Test");
106.     WNDCLASS wndclass;
107.     wndclass.style      = CS_HREDRAW|CS_VREDRAW ;
108.     wndclass.lpfnWndProc = WndProc ;
109.     wndclass.cbClsExtra  = 0 ;
110.     wndclass.cbWndExtra  = 0 ;
111.     wndclass.hInstance  = hInstance ;
112.     wndclass.hIcon       = LoadIcon (NULL, IDI_APPLICATION) ;
113.     wndclass.hCursor     = LoadCursor (NULL, IDC_ARROW) ;
114.     wndclass.hbrBackground = (HBRUSH) GetStockObject (WHITE_BRUSH)
        ;
115.     wndclass.lpszMenuName = NULL ;
116.     wndclass.lpszClassName = szAppName ;
117.
118.     // Register the window
119.     if(!RegisterClass(& wndclass))
120.     {
121.         MessageBoxA(NULL,"Registering the class failed","Error",MB
            _OK|MB_ICONERROR);
122.         exit(0);
123.     }
124.
125.     // CreateWindow
126.     HWND hwnd=CreateWindow(szAppName,TEXT("SetPixel example - code
        withc.com"),
127.                             WS_OVERLAPPEDWINDOW,
128.                             CW_USEDEFAULT,

```

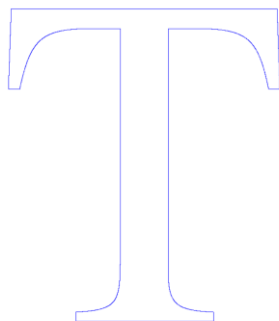
```

129.                                     CW_USEDEFAULT,
130.                                     CW_USEDEFAULT,
131.                                     CW_USEDEFAULT,
132.                                     NULL,
133.                                     NULL,
134.                                     hInstance,
135.                                     NULL);
136.     if(!hwnd)
137.     {
138.         MessageBoxA(NULL,"Window Creation Failed!","Error",MB_OK);
139.         exit(0);
140.     }
141.
142.     // ShowWindow and UpdateWindow
143.     ShowWindow(hwnd,iCmdShow);
144.     UpdateWindow(hwnd);
145.
146.     // Message Loop
147.     MSG msg;
148.     while(GetMessage(& msg,NULL,0,0))
149.     {
150.         TranslateMessage(& msg);
151.         DispatchMessage(& msg);
152.     }
153.
154.     return 0;
155. }

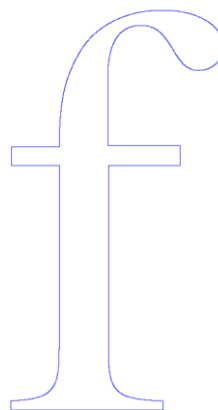
```

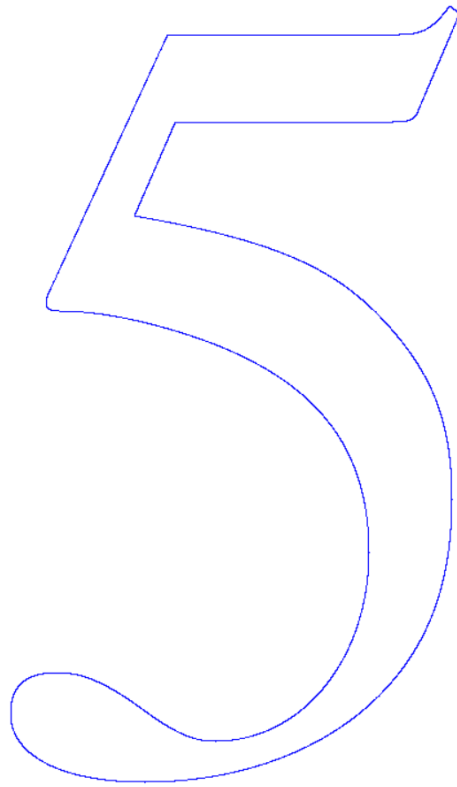
实验结果

T:



F





实验二

一、实验名称

使用 PageRank 算法计算网页的权重

二、实验原理

三、算法描述

1. 选择初始向量

$$x_0 = \left[\frac{1}{n} \quad \cdots \quad \frac{1}{n} \right]$$

2. 设置 $i = 1$

3. 计算 X_{i+1}

$$X_{i+1} = MX_i.$$

4. 分别计算 $X_{i+1} = \frac{X_{i+1}}{\|X_{i+1}\|}$ 和 $X_i = \frac{X_i}{\|X_i\|}$

$$\|X\| = \sqrt{\sum_{i=0}^n x_i}$$

5. 重复 3, 4 直到 $\text{error}(X'_{i+1} - X'_i) < \varepsilon$ i.e. $\|X'_{i+1} - X'_i\| < \varepsilon$

四、实验数据

输入的邻接矩阵

```
0.00 0.33 0.00 0.50
0.00 0.00 1.00 0.00
1.00 0.33 0.00 0.50
0.00 0.33 0.00 0.00
```

五、实验环境

OS: Windows 10

IDE: VScode

六、实验代码与结果

代码

```
1. #include "iostream"
2. #include "fstream"
3. #include "math.h"
4. using namespace std;
5. const int N = 200;
6.
7.
8. typedef struct web
```

```

9. {
10.     double web_m[N][N];
11.     int n;
12. }Web;
13.
14. typedef struct vec
15. {
16.     double vect[N];
17.     int n;
18.
19.     double norm () {
20.         double sum = 0;
21.         for (int i = 0; i < n; i++) {
22.             sum += vect[i]*vect[i];
23.         }
24.         return sqrt(sum);
25.     }
26.
27.     double distance (vec v) {
28.         double sum = 0;
29.         for (int i = 0; i < n; i++) {
30.             sum += (vect[i]-v.vect[i])*(vect[i]-v.vect[i]);
31.         }
32.         return sqrt(sum);
33.     }
34.
35.     vec normarliaztion() {
36.         vec temp;
37.         temp.n = n;
38.         double nor = norm();
39.         for (int i = 0; i < n; i++) {
40.             temp.vect[i] = vect[i] / nor;
41.         }
42.         return temp;
43.     }
44.
45. }Vec;
46.
47.
48.
49. Web network;
50. Vec R;
51.
52. void init() {

```

```
53.     ifstream infile;
54.     infile.open("pagerank.dat");
55.
56.
57.     infile >> network.n;
58.     R.n = network.n;
59.
60.     for (int i = 0; i < R.n; i++) {
61.         R.vect[i] = 1.0/R.n;
62.     }
63.
64.     // 读取连接数
65.     for (int i = 0; i < network.n; i++) {
66.         for (int j = 0; j < network.n; j++ ) {
67.             infile >> network.web_m[i][j];
68.         }
69.     }
70.     // 读取每个节点的最大出度
71.     int temp[network.n];
72.     for (int i = 0; i < network.n; i++) {
73.         int cnt = 0;
74.         for (int j = 0; j < network.n; j++ ) {
75.             if (network.web_m[j][i] == 1) {
76.                 cnt +=1;
77.             }
78.         }
79.         temp[i] = cnt;
80.     }
81.     // 构造 M 矩阵
82.     for (int i = 0; i < network.n; i++) {
83.         for (int j = 0; j < network.n; j++ ) {
84.             network.web_m[j][i] /= temp[i];
85.         }
86.     }
87.
88.     infile.close();
89. }
90.
91. void print() {
92.     for (int i = 0; i < network.n; i++) {
93.         for (int j = 0; j < network.n; j++) {
94.             printf("%.21f ", network.web_m[i][j]);
95.         }
96.         printf("\n");
```

```

97.     }
98. }
99.
100. void print_v() {
101.     for (int i = 0; i < R.n; i++) {
102.         printf("%.31f ", R.vect[i]);
103.     }
104.     putchar('\n');
105. }
106.
107. Vec mul(Web n, Vec R) {
108.     Vec temp;
109.
110.     for (int i = 0; i < R.n; i++) {
111.         temp.vect[i] = 0;
112.         temp.n = R.n;
113.     }
114.
115.     for (int i = 0; i < network.n; i++) {
116.         for (int j = 0; j < network.n; j++) {
117.             temp.vect[i] += n.web_m[i][j]*R.vect[j];
118.         }
119.     }
120.     for (int i = 0; i < R.n; i++) {
121.         R.vect[i] = temp.vect[i];
122.     }
123.     return temp;
124.
125. }
126.
127. int main (int argc, char*argv[])
128. {
129.     init();
130.     print();
131.
132.     printf("_____ \n");
133.
134.     Vec prev;
135.     do {
136.         prev = R;
137.         R = mul(network, R);
138.     } while (R.normarliaztion().distance(prev.normarliaztion()) >=
        0.01);
139.

```

```
140.     print_v();
141.
142.     return 0;
143. }
```

实验结果:

```
(base) PS C:\Users\1729\Desktop\projects\helloworld> .\pagerank.exe
0.00 0.33 0.00 0.50
0.00 0.00 1.00 0.00
1.00 0.33 0.00 0.50
0.00 0.33 0.00 0.00

-----
The 0 node's weight 0.177
The 1 node's weight 0.353
The 2 node's weight 0.352
The 3 node's weight 0.118
```